

Question number	Answer	Additional guidance	Mark
9(a)	1 (children) : 2.7 (women) : 4.7 (men) OR 1 (children) : 3 (women) : 5 (men)	ACCEPT 0.4 : 1 : 1.8 0.2 : 0.6 : 1 ACCEPT values in different order if clearly shown which is which	(1)

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9(b)	An explanation that includes three of the following points: - because formation of {plaques / tubercles / granuloma} in (lungs / other organs) (1) - therefore insufficient oxygen for (aerobic) respiration (1) - causes (other) organ failure (1) - weakens immune system so (death from) {other infections / tumours} (1)	ACCEPT reduces {gas exchange / gas exchange surface} ACCEPT description e.g. less oxygen enters blood ACCEPT named organ (in a correct context) / heart attack ACCEPT opportunistic infections (other than {TB / HIV} with / without ref to weakened immune system)	(3)

Question number	Answer	Additional guidance	Mark
9(c)(i)	An explanation that includes the following points: • because the untreated macrophages did not release IL-1β (1) • and therefore any inhibition by infection could not be shown / to compare the effect of infection on IL-1β (1)	ACCEPT did not release much IL-1β ACCEPT so that the inhibition could be seen to show the effect of TB (on macrophages / IL-1β)	(2)

Question number	Answer	Additional guidance	Mark
9(c)(ii)	An answer that includes four of the following points: - the antibody attached to the plastic otherwise everything will be washed away (1) - antibodies used (stage 1 or 5) are specific to the IL-1B (1) - so that they are bound to the {antigen / IL-1B} (stage 1 or 5) (1) - a dye has to be attached to antibodies to make {reaction / antigen / complex} visible (1) - unattached {antibodies / antigen / molecules / dye} have to be washed away (so that only the antigen-antibody complexes remain) (1)	ACCEPT stain the {antigen / complex}	(4)

Question number	Answer	Additional guidance	Mark
9(c)(iii)	A description that includes four of the following points: • both types of bacteria cultured in the same culture conditions (1) • use {aseptic technique / description of a technique} (1) • credit {named / description of} equipment used (1) • {calibration / standard} curve used to determine concentration of bacteria (1) • credit {description of how growth rate would be determined / appropriate stats test named to compare growth rate} (1)	ACCEPT same concentration of bacteria named condition e.g. temperature, time e.g. light meter / light sensor / spectrophotometer / colorimeter / cross underneath the flask / shine a light through and measure {transmission / absorbance} ACCEPT description of how the light {absorbed / transmitted} is determined by the number of bacteria ecf if haemocytometer used ecf if haemocytometer used	(4)

Question number	Answer	Additional guidance	Mark
9(c)(iv)	• use in a vaccine (1)	ACCEPT used to develop an {inhibitor of / antibody to / enzyme to break down / chemical to breakdown} this protein used to develop a way of switching off the gene coding for the protein (in the bacteria)	(1)

